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Does corporate sustainability performance foster green innovation? Firm-level evidence

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Introduction: Green innovation has become a key driver of sustainable industrial development, especially in emerging economies facing increasing environmental regulations and financing pressures. This study aims to empirically examine the relationship between corporate ESG performance and green innovation capability in Chinese A-share listed firms.

Methods: Using panel data from 2010 to 2023, a two-way fixed-effects model is employed to analyze the direct impact of ESG performance on green innovation, as well as the mediating roles of financing constraints and information asymmetry. The study also explores the moderating role of green finance development and the heterogeneous effects across ownership types, regions, and industries.

Results: The results show that stronger ESG performance significantly enhances green innovation. Mechanism analysis reveals that this effect is primarily driven by alleviating financing constraints and reducing information asymmetry. Additionally, green finance development further strengthens this positive relationship.

Discussion: Heterogeneity analyses indicate that the effect is more pronounced for firms in eastern regions, state-owned enterprises, pollution-intensive sectors, and technology-intensive industries. The findings provide theoretical insights and policy-relevant evidence to promote green innovation and sustainable industrial upgrading in emerging markets.

KEYWORDS

ESG performance, financing constraints, green finance, green innovation, information asymmetry

1 Introduction

Environmental, Social, and Governance (ESG) standards are now widely used to assess corporate excellence and competitive advantage in modern markets (Jian et al., 2025; Halisçelik and Soytaş, 2019), shifting from a marginal concern to a central element of strategic decision-making, investment allocation, and policy design (Laokulrach, 2025). In China where “carbon neutrality” and “carbon peak” guide long-term policy ESG has become a critical driver of sustainable economic development and green transformation (Lo and Xiang, 2025), underpinned by a decade-long policy evolution that lends particular urgency to understanding how ESG performance shapes green innovation behavior.

China’s ESG framework began crystallizing in 2015–2016 with the first national green bond guidelines, followed by the CSRC’s semi-mandatory environmental disclosure requirements in 2018. In 2020, China declared its “dual carbon” goals, elevating ESG to a strategic state imperative (Liang, L. et al., 2025; Liang, K. et al., 2025), while the 14th Five-Year Plan embedded binding carbon intensity targets into national planning. The effectiveness of China’s policy-driven

institutional reforms in generating measurable environmental and economic outcomes has been well documented: quasi-experimental evidence shows that national civilized city programmes have produced significant improvements in both economic performance and environmental quality across Chinese cities (Yang et al., 2025), while China's air quality has shown measurable improvement in response to sustained environmental regulation (Yang et al., 2026). Wu et al. (2025) further confirmed that fiscal technology expenditure boosts low-carbon innovation through R&D stimulation and FinTech development, while (Zhao et al., 2022) showed that carbon trading policy lowers regional energy intensity through industrial restructuring. By 2024, China's three major stock exchanges — under the unified guidance of the China Securities Regulatory Commission — issued parallel sustainability reporting guidelines (CSRC, 2024), and the 15th Five-Year Plan (2026–2030) now sets a 3.8% annual carbon intensity reduction target with dual control over total emissions making ESG disclosure and green innovation fundamental corporate imperatives. Firms failing to meet these standards face restricted financing and valuation discounts (Orbetsov, 2025).

Critically, ESG effectiveness in China operates through mechanisms distinct from Western literature. First, the dominance of state-owned enterprises (SOEs) which enjoy implicit government guarantees and preferential green credit access means the ESG–financing constraint channel operates differently across ownership types. Second, China's Confucian cultural heritage shapes reputation mechanisms: concepts such as *mianzi* and *guanxi* amplify ESG-linked reputational gains through dense relational networks while making reputational losses disproportionately consequential (Wan and Li, 2025). Third, China's policy-guided ESG evaluation frameworks are explicitly aligned with national development objectives (Wu and Zhu, 2025) rather than purely market-driven investor preferences making this a theoretically distinct setting that existing Western-oriented studies cannot adequately address.

Against this backdrop, corporate environmental innovation has become a foundation of sustainable economic growth (Wang and Li, 2024; Zu et al., 2026), yet traditional studies emphasized R&D investment, workforce characteristics, and competitive pressure (Taherdoost, 2024). ESG factors have since become a crucial analytical lens for explaining sustainable innovation strategies (Mukhtar et al., 2024a), as strong ESG performance enhances reputation, reduces financial and operational risks, and promotes innovation through improved resource allocation (Cabaleiro-Cerviño and Mendi, 2024). Recent firm-level evidence further confirms that ESG performance significantly enhances both the quantity and quality of green innovation by alleviating financing constraints and improving internal governance and innovation efficiency (Liang, L. et al., 2025; Liang, K. et al., 2025).

Three research gaps motivate this study. First, most studies treat ESG as a composite whole rather than examining sub-dimensions independently, producing inconsistent estimates and limiting explanatory power (Shi and Yao, 2025). Second, no unified theoretical framework explains the mediating and moderating conditions linking ESG to innovation; quantile regression analyses confirm the relationship varies substantially across firms, meaning average-effect models yield contradictory conclusions (Yuan et al., 2026). Third, China's unique market frictions including segmented capital markets, heterogeneous regulatory enforcement, SOE dominance, and Confucian relational governance are rarely accounted for in existing models. Moreover, the rapid expansion of urban digital infrastructure has emerged as an additional contextual factor reshaping ESG–innovation dynamics in

China, as digital construction has been shown to significantly stimulate economic growth and information flows across cities (Yang et al., 2024b), potentially amplifying the information asymmetry reduction channel through which ESG performance affects green innovation.

To address these gaps, this study pursues three explicit objectives. First, it empirically examines the direct effect of corporate ESG performance on green innovation capability using Chinese A-share listed firm data from 2010 to 2023. Second, it investigates the mediating mechanisms through which ESG performance influences green innovation specifically the roles of financing constraints and information asymmetry. Third, it explores the moderating role of green finance development and the heterogeneous effects of ESG across ownership structures, regional locations, and industry characteristics. To achieve these objectives, the study integrates stakeholder theory, information asymmetry theory, and resource-based theory in a demand–transmission–outcome framework, employing two-way fixed effects, mediation and moderation tests, and heterogeneity analysis. This framework is preferred over legitimacy theory, which focuses on symbolic ESG adoption, and institutional theory, which lacks firm-level micro-mechanisms. Theoretically, the study enriches interdisciplinary sustainability research; methodologically, it incorporates multiple mediating and moderating models; and practically, it provides policy-relevant evidence for strengthening green innovation incentives and ESG disclosure systems in China's emerging market context.

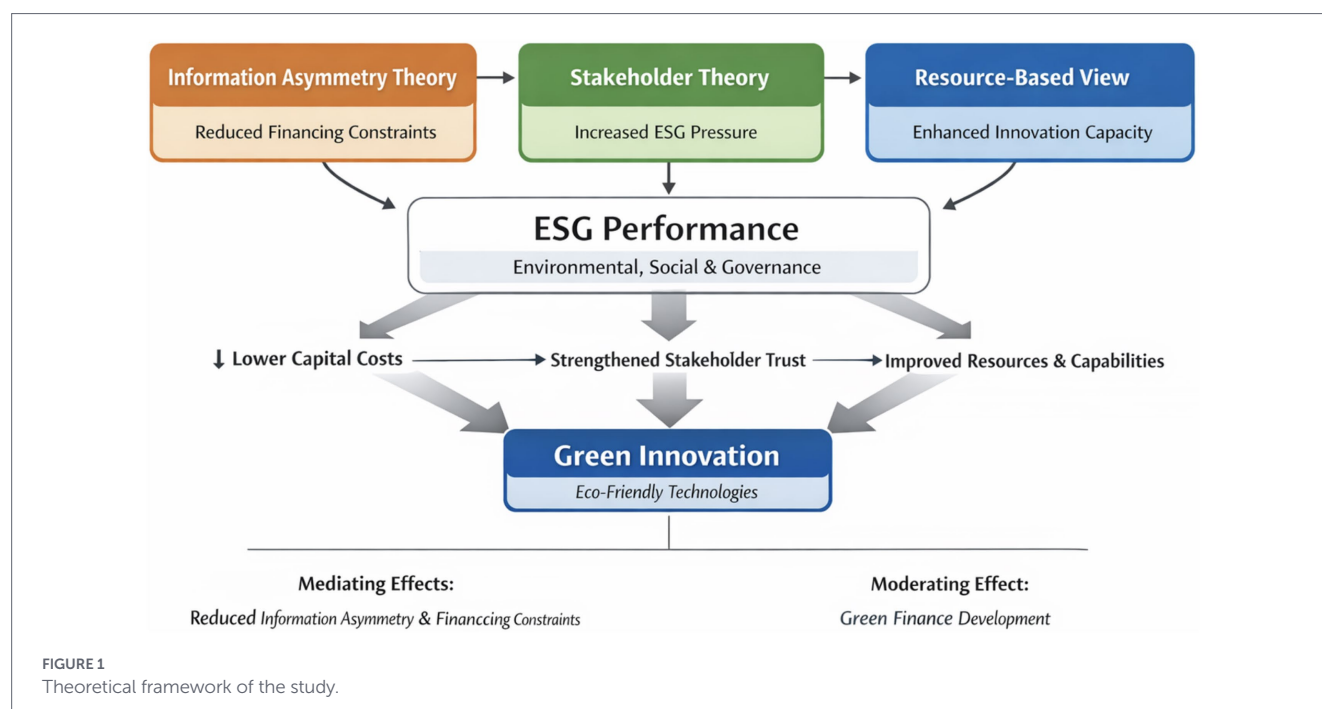
2 Theoretical framework

This study integrates three complementary theoretical pillars stakeholder theory, information asymmetry theory, and resource-based theory arranged in a sequential demand–transmission–outcome logic to explain how corporate ESG performance fosters green innovation (Figure 1). Stakeholder theory provides the demand-side motivation. Multi-party pressure from investors, regulators, communities, and supply chain partners compels firms to adopt credible ESG commitments as a strategic necessity for maintaining market access and long-term competitiveness. In China's institutional context, this pressure has been amplified by progressive mandatory disclosure requirements and binding carbon management targets, making ESG engagement a fundamental corporate imperative (Liang, L. et al., 2025; Liang, K. et al., 2025).

Information asymmetry theory identifies the transmission mechanism. Strong ESG performance credibly signals firm quality to capital markets, reducing information gaps, lowering financing costs, and easing access to green credit and long-term investment capital (Wang, 2025). This relaxation of financing constraints directly enables firms to fund the sustained R&D investment that green innovation requires.

Resource-based theory explains the supply-side outcome. The financial, human capital, and reputational resources unlocked through improved financing access are channeled into green innovation activity, generating the technological capabilities needed for environmentally oriented production (Abbas and Najam, 2025). This also explains the heterogeneous effects observed across ownership types, regions, and industries.

This framework was selected over legitimacy theory which focuses on symbolic ESG adoption rather than substantive financing dynamics and institutional theory which operates at the macro level without providing firm-level micro-mechanisms. Additionally, green finance



development moderates this overall process by amplifying the ESG–financing transmission channel, strengthening the positive effect of ESG performance on green innovation outcomes.

3 Review of the literature

3.1 Research on the economic impact of ESG performance

Academic research has widely examined how ESG attributes affect corporate financial behavior, offering insights into financing, investment efficiency, risk reduction, and innovation (Chen et al., 2023). As China advances its “dual carbon” goals, ESG has become a central element of non-financial disclosure and a key driver of long-term competitiveness (Wan and Li, 2025). Strong ESG performance improves capital access by reducing information gaps, boosting investor confidence, and facilitating equity and bond financing (Castillo, 2015). Under green finance policies, high-ESG firms receive preferential green credit and more favorable supplier terms that ease liquidity pressure (Eccles et al., 2014), while also attracting long-term investors toward green technologies and sustainable production (Voß et al., 2025). At the organizational level, ESG enhances productivity, reduces risk, and increases firm value (Aydoğmuş et al., 2022), lowers transaction costs through stronger stakeholder relationships (Prodanova and Tarasova, 2024), and reduces managerial short-termism through improved governance and transparency (Kim and Yoon, 2023). However, some studies warn that ESG assessments may incentivize greenwashing, where firms overstate responsibility through selective disclosure, potentially misleading investors and the public.

3.2 Research on the influencing factors of corporate green innovation

Green innovation combining environmental benefits with technological progress has become a key research focus (Islam, 2025),

with drivers viewed from two angles: external policy pressure and internal firm characteristics (Balasubramanian et al., 2021). On the policy side, environmental regulations, green finance, and low-carbon initiatives play major roles (Zhang, 2023). Moderate regulation stimulates green R&D, while overly strict rules raise compliance costs and hinder innovation (Gillan et al., 2021; Wu et al., 2022). Green finance channels capital toward sustainable technologies and combined low-carbon city pilots and green finance reforms significantly boost environmental innovation through green industry clustering and stricter supervision. Environmental inspections and carbon trading further support green innovation (Kim and Park, 2023). Internally, environmentally conscious executives embed green goals into corporate strategy, digital transformation enhances information flows and technological capability (Wei et al., 2023), and governance factors such as board independence and ownership structure promote long-term sustainability-oriented R&D (Xie et al., 2017).

3.3 Research on the impact of ESG performance on green innovation

Research on ESG and green innovation emphasizes two key perspectives: stakeholder theory and signaling theory. Strong ESG performance signals sustainability commitment, attracting eco-conscious investors and securing funding for green projects (Hu et al., 2021), while building trust with governments, communities, and consumers (Fredriksson and Millimet, 2002). The main transmission channels are financing constraints, information asymmetry, and managerial motivation. Strong ESG practices particularly boost environmental innovation in state-owned enterprises and regions with strong intellectual property protections (Wan et al., 2024). ESG facilitates green development by lowering borrowing costs, reducing short-termism, and optimizing supply chain funding (Gao et al., 2025), while green finance amplifies ESG’s impact through preferential lending and shared risk frameworks (Zhang, 2025). Despite these positive effects, gaps remain: many studies treat ESG, neglecting the

distinct roles of its environmental, social, and governance components, and the variation in ESG's impact across firm stages, industries, and locations.

4 Research hypotheses

This study integrates information asymmetry, stakeholder interests, and resource management to examine how ESG performance fosters environmental innovation (Agyemang et al., 2025). As a core non-financial indicator, ESG improves external financing through signaling mechanisms and enhances innovation efficiency through internal governance optimization (Zhao et al., 2025), with green finance acting as an external institutional force that amplifies these effects.

4.1 ESG performance and corporate green innovation

Strong ESG performance influences green innovation through both promotional and inhibitory pathways. Promotionally, ESG signals sustainability commitment, boosts investor confidence, reduces financing costs (Tariq et al., 2017), and attracts stakeholder support. Regulatory pressures further encourage firms to integrate green responsibilities into strategy and increase R&D investment (Kim and Park, 2023), while stakeholder theory suggests that prioritizing ESG standards prompts external groups to drive technological advancement and foster long-term relationships (Fredriksson and Millimet, 2002). However, ESG can also inhibit innovation: disclosure and compliance costs divert resources, and greenwashing where firms improve ratings without substantive action initially boosts but later suppresses innovation through an inverted U-shaped effect, ultimately misallocating resources and stifling genuine technological breakthroughs (Gao et al., 2025). Therefore, we propose:

H1a: ESG performance substantially enhances corporate green innovation.

H1b: ESG performance may inhibit corporate green innovation.

4.2 The mediating mechanism of financing constraints

Green innovation is characterized by significant risks and long development periods, making it particularly vulnerable to financing constraints (Cao et al., 2021). ESG performance alleviates these limitations through three pathways: enhancing credit ratings and reducing debt financing costs (Tan and Zhu, 2022); attracting green-oriented investors and expanding equity financing opportunities (Taheri Hosseinkhani, 2025); and strengthening supply chain trust, facilitating access to trade credit from suppliers willing to offer flexible payment terms (Eccles et al., 2014). By easing these constraints, ESG enables firms to secure funding for environmental projects (Zhai et al., 2022) and advance R&D in eco-friendly technologies that require significant upfront capital (Zhou et al., 2021). Therefore, we propose:

H2: ESG performance promotes corporate green innovation by reducing financing constraints.

4.3 The mediating mechanism of information asymmetry

Information asymmetry is a major obstacle to green technology advancement (Wang et al., 2021). ESG disclosure mitigates this through three channels: improving corporate transparency and lowering investor risk assessment costs (Buallay, 2019); strengthening external oversight from media and institutional investors, curbing managerial short-termism and encouraging long-term green innovation (Yang et al., 2024a); and enhancing corporate governance to optimize green innovation decision-making (Xie et al., 2017). Therefore, we propose:

H3: ESG performance promotes corporate green innovation by reducing information asymmetry.

4.4 The moderating role of green finance

Green finance links ESG outcomes to sustainable development through regulatory frameworks and economic incentives (Junaedi, 2024). Commercial banks offer preferential loan rates to green enterprises reflecting enhanced credit profiles, stricter environmental regulations, and competition in the green credit market (Luo et al., 2021), while instruments like green insurance mitigate innovation risks and encourage green transformation (Hu et al., 2023). Overall, this pricing structure demonstrates market efficiency (Usai et al., 2021). As green finance develops, it provides a favorable institutional environment that eases financing constraints and facilitates the translation of ESG efforts into innovation outcomes (Lu et al., 2025). Therefore, we propose:

H4: Green finance amplifies the positive effect of ESG performance on corporate green innovation, the greater the green financial resources, the stronger the ESG-driven innovation.

4.5 Theoretical expectations of heterogeneous effects

China's institutional environment conditions the heterogeneous effects of ESG on green innovation in ways absent from Western contexts. The dominance of SOEs which access policy-directed green finance regardless of ESG rating quality means the financing constraint channel is more pronounced for private firms, whose ESG-driven innovation advantages derive more from compliance obligations and policy alignment (Zhang, 2023). Confucian cultural values further shape how ESG disclosures are interpreted: in regions where relational governance norms are stronger, ESG generates larger reputational returns through guanxi networks, amplifying innovation effects beyond market-based predictions (Wan and Li, 2025). Resource dependency theory further suggests that eastern regions with advanced green financial infrastructure and talent pools show more pronounced ESG-driven innovation effects (Fei et al., 2024; Do Carmo Vieira and Borges, 2025), while heavily polluting sectors face stricter compliance needs that intensify ESG's innovation role (Kim and Park, 2023), and technology-intensive firms more efficiently convert ESG resources into innovation outcomes through stronger R&D capabilities (Casciello et al., 2024). These heterogeneous effects highlight the complexity of ESG's role and provide the theoretical foundation for our empirical heterogeneity analysis.

5 Materials and methods

5.1 Model construction

5.1.1 Baseline regression model

To delve deeper into the effects of ESG performance on business sustainability initiatives, this research employs a two-way fixed effects methodology for empirical examination. To test research hypothesis H1, the following model is developed:

$$GIN_{it} = \alpha_1 + \beta_1 ESG_{it} + \theta_1 \times X_{it} + \omega_i + \varphi_t + \varepsilon_{it} \quad (1)$$

where the subscript i denotes the firm, and t denotes the year. GIN_{it} is the dependent variable, representing the green innovation of firm i in year t . X_{it} represents a set of control variables. ω_i denotes the firm fixed effects. φ_t shows the time fixed effects and ε_{it} is the random disturbance term affecting corporate green innovation. The coefficient β_1 is the primary focus of this study, reflecting the direct impact of corporate ESG on its green technology innovation.

5.1.2 Mediating effect model

To examine whether ESG performance influences corporate green innovation through mediating mechanisms and following the methodological approach of (Jiang, 2022) and (Amore and Bannedsen, 2016), we develop a mediating effect model building on Equation 1. The specific models are as follows:

$$Mediation_{it} = \alpha_1 + \beta_1 ESG_{it} + \theta_1 X_{it} + \omega_i + \varphi_t + \varepsilon_{it} \quad (2)$$

$$GIN_{it} = \alpha_2 + \beta_2 Mediation_{it} + \theta_2 X_{it} + \omega_i + \varphi_t + \varepsilon_{it} \quad (3)$$

where the variables in Equations 2, 3 retain their definitions from Equation 1, and Mediation represents the mediating variable.

5.1.3 Moderating effect model

To see whether the relationship between ESG performance and the company's green innovation will be affected by other factors, we built a moderating effect model based on Equation 1. The specific model is as follows:

$$EC_{it} = \alpha_3 + \beta_1 ESG_{it} + \beta_2 Moderator_{it} + \beta_3 ESG_{it} \times Moderator_{it} + \theta_1 X_{it} + \delta_i + \sigma_t + \mu_{it} \quad (4)$$

where the variables in Equation 4 retained the same definitions from Equation 1, and Moderator represents the moderating variable.

5.2 Variable measurement

Dependent variable: This study measures corporate green innovation using the number of green patent applications, preferred over patent grants due to administrative delays and time lags. We distinguish between two types: substantive green innovation measured by green invention patents representing significant technological breakthroughs and strategic green innovation measured by green utility

model patents reflecting minor practical improvements. Both are quantified as the natural logarithm of total applications plus one.

Explanatory variable: Corporate ESG performance is measured using the Sino-Sec ESG rating system, which incorporates both international frameworks and China-specific conditions. The nine rating categories (C to AAA) are converted to numerical scores from 1 to 9, with higher scores reflecting superior ESG performance.

Mediating variables: Financing constraints are measured using the SA index, where a higher absolute value indicates more severe financial difficulty. Information asymmetry is measured using three stock liquidity indicators liquidity ratio (LR), illiquidity ratio (ILL), and gamma inversion (GAM) following (Lee et al., 2022) and (Huang et al., 2022). Poorer stock liquidity signals greater information asymmetry. Principal component analysis is applied to these three indicators to construct a comprehensive information asymmetry index (ASY).

Moderating variable: Green finance is measured at the city level using an entropy-based index constructed following (Zhou et al., 2022), capturing four dimensions: eco-friendly lending, green capital market instruments, sustainable investment vehicles, and environmentally responsible insurance coverage.

Control variables: We control for firm size (natural log of total assets), asset-to-liability ratio, return on total assets, cash flow ratio, fixed asset ratio, operating revenue growth rate, financial leverage, board size (natural log of total directors), proportion of independent directors, top ten shareholders' shareholding ratio, and firm age (natural log of years since establishment plus one).

5.3 Data source and descriptive statistics

The sample comprises Chinese A-share listed firms over the period 2010–2023. Firms designated as ST, *ST, or PT, those operating in the financial and real estate sectors, and observations with substantial missing values were excluded. All continuous variables were winsorized at the 1st and 99th percentiles to mitigate the influence of extreme values, yielding a final dataset of 25,620 firm-year observations. ESG performance data were sourced from Wind and Bloomberg, while green innovation data were obtained from the China Research Data Service Platform (CNRDS), which is particularly well-suited to the Chinese institutional context as it integrates industry, government, and firm-level information under a regulatory environment characterized by government guidance and state-owned enterprise leadership. Supplementary financial, governance, and industry-level data were collected from the China Stock Market and Accounting Research (CSMAR) database, the China Statistical Yearbook, and the China Industrial Statistical Yearbook.

Descriptive statistics are reported in Table 1. The green innovation variable exhibits considerable cross-firm variation, with a mean of 0.380 and standard deviation of 0.781, reflecting heterogeneous innovative capacity across the sample. ESG scores range from 1 to 8 with a mean of 4.206, indicating substantial dispersion in corporate sustainability performance. Financial health indicators reveal that sampled firms maintain moderate leverage (0.416), modest profitability as measured by return on assets (0.037), and stable cash flow ratios (0.047), with average revenue growth of 15.1%. Governance characteristics indicate boards of moderate size with an independent director ratio averaging 0.375, while ownership concentration among the top ten shareholders averages 0.581, suggesting relatively concentrated equity structures typical of Chinese listed firms.

6 Results and discussion

6.1 Impact of ESG performance on green innovation

Table 2 reports the baseline regression results examining the effect of ESG performance on corporate green innovation. Column (1) presents unadjusted estimates, while Column (2) incorporates all control variables along with firm and year fixed effects. In both specifications, the ESG coefficient remains significantly positive at the 1% level, confirming H1a and demonstrating that stronger ESG performance consistently promotes green innovation. The result further aligns with

(Mukhtar et al., 2024b), who showed that strong ESG performance enhances organizational learning and resource allocation in ways that directly stimulate environmental innovation. Theoretically, these results support the resource-based view: comprehensive ESG practices through environmental responsibility, social engagement, and stronger governance improve firms' resource efficiency and stimulate green technology development by unlocking financing access and reputational capital that would otherwise constrain innovation investment.

Columns (3) and (4) further distinguish between substantive green innovation and strategic green innovation, with ESG performance showing a significantly positive effect on both types at the 1%

TABLE 1 Descriptive statistics of basic variables.

Variable	N	Mean	Standard deviation	Minimum	Maximum
GIN	25,620	0.380	0.781	0	3.871
ESG	25,620	4.206	0.901	1	8
Size	25,620	22.146	1.242	19.525	26.452
Lev	25,620	0.416	0.204	0.028	0.908
ROA	25,620	0.037	0.064	−0.556	0.222
CashFlow	25,620	0.047	0.069	−0.222	0.267
Fixed	25,620	0.215	0.157	0.001	0.734
Growth	25,620	0.151	0.401	−0.673	5.076
FL	25,620	1.254	1.002	−2.022	11.556
Board	25,620	2.119	0.198	0	2.708
Indep	25,620	0.375	0.053	0	0.600
Top10	25,620	0.581	0.152	0.195	0.910
FirmAge	25,620	2.944	0.349	0	4.304

TABLE 2 Baseline regression.

Variable	(1) GIN	(2) GIN	(3) GIN	(4) GIN1
ESG	0.0553*** (0.0054)	0.0533*** (0.0055)	0.0279*** (0.0045)	0.0441*** (0.0034)
Size		0.0455*** (0.0132)	0.0434*** (0.0106)	0.0179*** (0.0063)
Lev		0.0380 (0.0415)	0.0285 (0.0334)	0.0285 (0.0259)
ROA		0.1165* (0.0642)	0.0732 (0.0532)	0.0818 (0.0518)
CashFlow		−0.1175** (0.0575)	−0.0660 (0.0462)	−0.0735* (0.0432)
Fixed		0.0186 (0.0541)	0.0013 (0.0424)	0.0289 (0.0327)
Growth		−0.0116* (0.0065)	−0.0079 (0.0052)	−0.0105* (0.0061)
FL		0.0027 (0.0034)	0.0031 (0.0029)	0.0005 (0.0026)
Board		0.0072 (0.0526)	−0.0257 (0.0411)	0.0172 (0.0260)
Indep		0.1501 (0.1714)	0.0748 (0.1364)	0.0771 (0.0811)
Top10		−0.1682*** (0.0607)	−0.1167** (0.0493)	−0.0814*** (0.0321)
FirmAge		0.1393 (0.1126)	0.1029 (0.0905)	0.1340*** (0.0390)
Constant	−0.0456 (0.0308)	−1.3367*** (0.4343)	−1.1050*** (0.3406)	−0.8092*** (0.1724)
FirmFE	YES	YES	YES	YES
YearFE	YES	YES	YES	YES
R ²	0.0367	0.0406	0.0329	0.0225

*, **, and *** denote statistical significance of parameters at the 10, 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

level. This pattern also corroborates (Long et al., 2023), whose quantile regression analysis demonstrated that ESG's effect on green innovation is heterogeneous across the innovation distribution, with firms at lower innovation levels responding more strongly through strategic rather than substantive innovation channels. Overall, these results confirm that better ESG performance enhances environmental awareness, corporate reputation, and financing access, enabling firms to expand both the depth and breadth of their green innovation activities.

Among the control variables, firm size is positively associated with green innovation. The negative and significant coefficient on shareholding concentration suggests that concentrated ownership may entrench managerial conservatism and reduce innovation incentives, consistent with corporate governance literature showing that ownership concentration can impede long-term investment in uncertain activities such as green R&D (Khan et al., 2025).

6.2 Robustness tests

6.2.1 Addressing confounding effects of other policies

To address potential distortions from the “Smart City” initiative and the “China Carbon Emission Trading Pilot” programme, two policy dummy variables are incorporated into the baseline model. As shown in Columns (1) and (2) of Table 3, the ESG coefficient remains significantly positive at the 1% level after controlling for both policy effects, consistent with the baseline estimates. This confirms that the positive impact of ESG performance on corporate green innovation is robust and not driven by these concurrent environmental or innovation policies, which is consistent with (Wang et al., 2025a), who similarly demonstrated that ESG-driven innovation effects persist independently of carbon market policy interventions, providing additional support for H1a.

6.2.2 Replacing the dependent variable

To strengthen the robustness of the findings, the measure of green innovation is replaced from patent applications to granted patents, and the regression model is re-estimated. Columns (1) and (2) of Table 4 report the results with and without control variables, respectively. The ESG coefficients remain significantly positive at the 1%

level in both specifications, indicating that firms with stronger ESG performance continue to exhibit higher levels of green technological innovation even under this more stringent innovation measure. This approach is consistent with (Long et al., 2023), who similarly validated the robustness of ESG–green innovation findings by alternating between patent application and grant-based indicators among Chinese A-share listed firms, thereby reaffirming the baseline conclusion and providing further support for H1a.

6.2.3 Changing the clustering level

To test the sensitivity of our baseline results to different clustering methods, the regression model is re-estimated using city-level clustered standard errors instead of firm-level clustering. Columns (1) and (2) of Table 5 report the results without and with control variables, respectively. The ESG coefficient remains closely aligned with the baseline estimates and retains its significance at the 1% level, indicating that even after accounting for potential intra-city correlations, the positive effect of ESG performance on corporate green innovation remains robust. This finding is consistent with (Wang et al., 2025b), who demonstrated that ESG-driven green innovation effects remain stable across alternative clustering specifications in Chinese listed firm samples, providing additional support for H1a.

6.2.4 Winsorization

To address potential bias from extreme values, the main dependent variable—corporate green technology innovation (GIN)—is

TABLE 4 Replacing the dependent variable.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0051*** (0.0010)	0.0051*** (0.0010)
Constant	0.0106** (0.0053)	−0.0602 (0.0728)
Control	NO	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0055	0.0063

** and *** denote statistical significance of parameters at the 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

TABLE 3 Addressing confounding effects of other policies.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0533*** (0.00545)	0.0534*** (0.00545)
Addressing confounding effects of other policies	0.0106 (0.0200)	−0.0494** (0.0236)
Constant	−1.343*** (0.435)	−1.351*** (0.434)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.041	0.041

** and *** denote statistical significance of parameters at the 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

TABLE 5 Relationship between ESG performance and green innovation with and without controls.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0553*** (0.0061)	0.0533*** (0.0061)
Constant	−0.0456 (0.0369)	−1.3367** (0.5187)
Control	NO	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0367	0.0406

** and *** denote statistical significance of parameters at the 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

TABLE 6 Results of winsorization.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0492*** (0.0052)	0.0403*** (0.0039)
Constant	−1.0005** (0.3913)	−0.6658** (0.2841)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	3,485	3,426

** and *** denote statistical significance of parameters at the 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

TABLE 7 Shortening the sample period.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0352*** (0.0049)	0.0341*** (0.0050)
Constant	−0.0250 (0.0248)	−0.5047 (0.3182)
Control	NO	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0259	0.0297

***denote statistical significance of parameters at the 1%. The given values in parentheses present adjusted standard errors.

winsorized at the 1 and 5% levels and the model is re-estimated. Columns (1) and (2) of Table 6 report the results. Under both winsorization settings, the ESG coefficients remain highly significant at the 1% level and are consistent in both magnitude and direction with the baseline findings, indicating that the positive effect of ESG performance on corporate green innovation is not driven by outliers. This is consistent with (Wang et al., 2025a), who similarly confirmed the stability of ESG–green innovation estimates under alternative winsorization thresholds among Chinese A-share listed firms, thereby reinforcing H1a and confirming that the original conclusions are robust to the influence of extreme observations.

6.2.5 Shortening the sample period

To assess whether the baseline results were affected by the external shock of COVID-19, the sample is restricted to the pre-pandemic period (2019–2022) and the fixed-effects model is re-estimated. Columns (1) and (2) of Table 7 report the outcomes without and with control variables, respectively. The ESG coefficients remain positive and significant at the 1% level in both specifications, indicating that ESG performance continues to promote corporate green innovation even after removing potential pandemic-related distortions. This finding validates H1a and demonstrating that the baseline conclusions are not driven by COVID-19-related structural breaks.

6.2.6 Instrumental variable

To address potential endogeneity arising from omitted variables or reverse causality, an instrumental variable (IV) approach is

employed. Following peer-effect theory, the average ESG performance of Chinese securities firms in the same year, region, and industry is used as the IV for firm-level ESG. This instrument satisfies the relevance requirement because corporate ESG behaviors tend to cluster within industries and regions due to shared norms and policy environments and meets the exogeneity requirement as industry-level average ESG affects a firm's green innovation only indirectly through its influence on the firm's own ESG choices.

The 2SLS estimation results confirm the validity of the IV strategy. In the first stage, the IV coefficient is strongly significant at the 1% level. The Kleibergen-Paap rk LM statistic rejects underidentification, and the Kleibergen-Paap rk Wald F statistic exceeds the 10% maximal-bias threshold of 16.38, eliminating concerns about weak instruments. In the second stage, the ESG coefficient remains significantly positive at the 1% level and is consistent in direction with the baseline results, as reported in Table 8. Overall, the IV analysis confirms a causal relationship between ESG performance and green innovation, providing strong support for H1a and establishing a solid foundation for evaluating the mediation mechanisms related to financing constraints (H2) and information asymmetry (H3).

6.3 Mechanism analysis

6.3.1 Mediating mechanism of financing constraints

This section examines how ESG performance promotes corporate green innovation through the alleviation of financing constraints. Using stepwise regression, Column (1) of Table 9 shows that ESG performance is significantly negatively associated with financing constraints at the 1% level, indicating that stronger ESG practices help relieve financial pressure on firms. Column (2) further reveals that financing constraints are significantly negatively related to green innovation, suggesting that firms facing fewer financial hurdles are better positioned to invest in green innovation activities. Together, these results confirm the mediating pathway—ESG performance alleviates financing constraints, which in turn enhances green innovation, providing empirical support for H2.

These findings align with the theoretical mechanisms proposed by (Li et al., 2025), who demonstrated that strong ESG performance improves corporate credit ratings and reduces debt financing costs, attracts green-oriented institutional investors to expand equity financing channels, and strengthens supply chain relationships to secure more favorable trade credit terms. These channels collectively enhance capital allocation efficiency and provide the sustained financial support that green innovation, characterized by long R&D cycles, high uncertainty, and significant upfront capital requirements demands. The empirical evidence therefore confirms that ESG performance serves as a critical mechanism for advancing green technological progress through the easing of financing constraints.

6.3.2 Mediating mechanism of information asymmetry

To examine how ESG performance promotes green technology innovation through reduced information asymmetry, a mediating effect model is employed. As shown in Table 10, ESG performance significantly lowers information asymmetry, indicating that stronger ESG practices improve corporate transparency and enhance external monitoring. Information asymmetry in turn is negatively associated

TABLE 8 Results of instrumental variable test.

Variable	(1) The First Stage	(2) The Second Stage
IV1	0.9158*** (14.2855)	
Kleibergen-PaaprkmStatistic	119.402 (0.0000)	
Kleibergen-PaaprkmWaldFStatistic	204.085 (16.38)	
Constant	−4.3740*** (−7.4880)	
Control	YES	YES
R ²	0.509	−0.015
FirmFE	Yes	Yes
YearFE	Yes	Yes

***denote statistical significance of parameters at the 1%. The given values in parentheses present adjusted standard errors.

TABLE 9 Financing constraints.

Variable	(1) Mediation	(2) GIN
ESG	−0.0024*** (0.0008)	
Mediation		−0.3003*** (0.1035)
Constant	3.4651*** (0.0990)	−0.3424 (0.4822)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.8866	0.0355

***denote statistical significance of parameters at the 1%. The given values in parentheses present adjusted standard errors.

TABLE 10 Mediating mechanism of information asymmetry.

Variable	(1) Mediation	(2) GIN
ESG	−0.0070** (0.0035)	
Mediation		−0.0488*** (0.0141)
Constant	5.7123*** (0.2553)	−1.1044** (0.4290)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.5179	0.0352

** and *** denote statistical significance of parameters at the 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

with green innovation, meaning that firms operating in better information environments generate higher levels of green innovation activity. These results provide empirical support for H3.

These findings are consistent with (Ramadhan, 2025), who demonstrated that ESG disclosure mitigates information asymmetry by improving corporate transparency, strengthening governance structures, and creating reliable sustainability signals for capital markets.

Together, these channels reduce information frictions that would otherwise impede the allocation of capital toward green innovation projects. The empirical evidence therefore confirms that ESG performance not only improves financing conditions through signaling mechanisms but also reduces information barriers, providing important micro-level insights into how ESG factors shape corporate innovation behavior.

6.3.3 The moderating mechanism of green finance

To test whether green finance strengthens the effect of ESG performance on corporate green innovation (H4), an interaction term is introduced into the model. To prevent structural multicollinearity between the interaction term and its constituent variables a concern raised by preliminary estimations, a mean-centering procedure is applied following (Aiken et al., 1991), subtracting the sample mean from both the ESG score and the green finance index before generating their interaction term. VIF diagnostics confirm that multicollinearity is not a concern in the re-estimated model.

As shown in Table 11, the main effect of ESG, representing its impact on green innovation when green finance is at the sample mean is significantly positive at the 1% level, and the interaction term is significantly positive at the 5% level, indicating that a well-developed green financial system amplifies the marginal impact of ESG performance on green innovation, providing robust empirical support for H4.

The theoretical mechanism behind this moderating effect is two-fold. First, green finance directs credit resources toward high-ESG firms, reducing the financing costs of green R&D through differentiated pricing (Usai et al., 2021). Second, financial instruments such as green insurance mitigate innovation risks, increasing firms' willingness to pursue long-term green technology development (Buallay, 2019). Together, these mechanisms operate within an institutional environment shaped by stricter environmental regulations and stronger competition in the green finance market, both of which enhance the credit profile and innovation incentives of green enterprises (Zhou et al., 2021), offering micro-level evidence of how institutional conditions interact with corporate sustainability behaviors to drive green technological progress.

6.4 Heterogeneity analysis

6.4.1 Regional attributes

Firms in different regional environments experience varying impacts of ESG performance on green technology innovation, consistent with Resource Dependence Theory. As shown in Table 12, ESG performance exerts a significantly positive effect on green innovation in both eastern and non-eastern regions, though the effect is stronger in eastern China. This regional difference is consistent with theoretical expectations that ESG performs more effectively in institutionally developed environments.

This disparity can be explained by regional differences in financing constraints and information asymmetry. Eastern regions benefit from advanced green financial systems and higher marketization, allowing firms to signal ESG performance more effectively, reducing financing costs for green R&D and enhancing the conversion of ESG resources into innovation outcomes (Tan and Zhu, 2022). In non-eastern regions, financing frictions limit this conversion efficiency.

TABLE 11 The moderating mechanism of green finance.

Variable	(1)
	GIN
c_ESG	0.0540*** (0.0055)
c_GF	0.6834*** (0.1349)
c_ESG*c_GF	0.1153** (0.0484)
Constant	−1.0568** (0.4309)
FirmFE	YES
YearFE	YES
R ²	0.0427

** and *** denote statistical significance of parameters at the 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

TABLE 12 Regional attributes.

Variable	(1)	(2)
	Eastern region	Non-eastern region
	GIN	GIN
ESG	0.0561*** (0.0072)	0.0491*** (0.0082)
GF	0.0517*** (0.0175)	0.0263 (0.0196)
ESG*GF	−0.0157 (0.0541)	0.1146* (0.0647)
Constant	−2.2780*** (0.5569)	0.2799 (0.6347)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0537	0.0291

* and *** denote statistical significance of parameters at the 10, and 1%, respectively. The given values in parentheses present adjusted standard errors.

Eastern regions also benefit from stronger external information environments including greater media coverage and a higher density of institutional investors which reduce information asymmetry and further support green innovation (Buallay, 2019). The concentration of universities and research institutions additionally facilitates knowledge spillovers and talent access (Fredriksson and Millimet, 2002).

Regarding the moderating role of green finance, a notable regional asymmetry emerges. In non-eastern regions, the ESG × GF interaction is significantly positive, indicating that green finance amplifies ESG's innovation effect through policy incentives and risk-mitigating instruments. In eastern regions, this interaction is insignificant, suggesting that mature financial systems reduce the marginal contribution of additional green finance development. These findings underscore the importance of region-specific policies to effectively promote corporate green transformation, particularly by expanding green financial infrastructure in central and western provinces where its marginal impact is greatest.

6.4.2 Ownership heterogeneity

Table 13 shows that ESG performance has a stronger positive impact on green innovation in state-owned enterprises (SOEs) than in non-state-owned enterprises (non-SOEs), with both coefficients significant at the 1% level. These results support the expectation that

TABLE 13 Ownership heterogeneity.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0601*** (0.0093)	0.0480*** (0.0067)
GF	0.0306 (0.0230)	0.0651*** (0.0161)
ESG*GF	0.0406 (0.0773)	0.0378 (0.0491)
Constant	−0.2608 (0.8009)	−2.1938*** (0.5157)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0535	0.0379

*** denote statistical significance of parameters at the 1%. The given values in parentheses present adjusted standard errors.

ESG performance has a more pronounced innovation-promoting effect in SOEs.

Two underlying mechanisms explain this difference. First, regarding financing constraints, SOEs benefit from implicit government backing and stronger institutional support, enabling them to translate ESG performance more efficiently into financing advantages. Their ESG activities are more likely to attract preferential loans from policy banks and government green subsidies (Xie et al., 2017), while their larger scale and stronger asset collateral capacity allow them to secure more favorable trade credit terms (Eccles et al., 2014). Non-SOEs, by contrast, face higher credit discrimination and risk premiums that limit the financing benefits of ESG improvement.

Second, through the information asymmetry channel, SOEs operate under stronger policy pressure and public scrutiny, resulting in more standardized and transparent ESG disclosures that enhance their signaling effectiveness and reduce information frictions between firms and investors (Buallay, 2019). Consistent with stakeholder theory, SOEs' sensitivity to government expectations also pushes them to prioritize political legitimacy and social reputation through strong ESG performance (Fredriksson and Millimet, 2002), further strengthening their strategic orientation toward green innovation. Together, these findings confirm that ownership structure shapes the magnitude of ESG's innovation effect, with SOEs deriving greater green innovation benefits from ESG engagement.

6.4.3 Heavy-polluting industry heterogeneity

To examine whether industry characteristics shape the ESG–green innovation relationship, firms are classified into heavy-polluting and non-polluting sectors based on the Guidelines for Industry Classification of Listed Companies (2012) issued by the China Securities Regulatory Commission. As shown in Table 14, ESG performance exerts a significantly positive effect on green innovation in both groups at the 1% level, though the effect is stronger for heavy-polluting enterprises. These results suggest that firms facing higher environmental burdens derive greater innovation benefits from ESG engagement.

This heterogeneity is consistent with resource dependence and stakeholder theories. Heavy-polluting enterprises operate under stricter environmental regulations and stronger external monitoring, which means strong ESG performance allows these firms to send more credible signals about environmental responsibility, reducing financing

TABLE 14 Heavy-polluting industry heterogeneity.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0579*** (0.0104)	0.0496*** (0.0064)
GF	0.0337 (0.0243)	0.0557*** (0.0159)
ESG*GF	−0.0026 (0.0713)	0.0411 (0.0508)
Constant	0.2117 (0.8641)	−1.8381*** (0.4944)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0342	0.0481

*** denote statistical significance of parameters at the 1%. The given values in parentheses present adjusted standard errors.

constraints and lowering the cost of capital for green R&D. By mitigating information asymmetry, ESG practices also enhance governance quality and curb managerial short-termism, encouraging firms to channel more resources into green technology development (Bualay, 2019). Furthermore, because heavily polluting firms face higher compliance costs and greater environmental risks, they have stronger incentives to pursue green transformation through improved ESG performance (Fredriksson and Millimet, 2002). In contrast, firms in cleaner industries encounter lighter regulatory pressure, and ESG-driven benefits may be more easily allocated to non-environmental initiatives, diminishing their direct impact on green innovation.

6.4.4 Labor heterogeneity

To explore whether the impact of ESG performance on green innovation varies by factor intensity, firms are divided into labor-intensive and non-labor-intensive groups using the median employee-to-total-assets ratio as the threshold. As shown in Table 15, ESG performance significantly promotes green innovation in both groups at the 1% level, though the effect is noticeably stronger among non-labor-intensive firms, whose ESG coefficient is nearly twice that of labor-intensive enterprises. This indicates that firms with lower reliance on labor derive greater innovation benefits from ESG engagement.

This heterogeneity can be explained through the resource-based view and the financing constraint mechanism. Non-labor-intensive firms typically depend more on technological capabilities and R&D to maintain competitiveness, and their higher fixed-asset ratios and stronger technological infrastructure provide a solid base for green innovation. Strong ESG performance further alleviates financing constraints through signaling, reducing the cost of capital and facilitating sustained investment in long-term green R&D (Ding et al., 2024). These firms also face greater technological complexity and higher information asymmetry, meaning ESG practices by improving governance and strengthening external oversight more effectively reduce managerial short-termism and enhance strategic commitment to green technology development (Wu et al., 2026).

6.5 Environmental, social, and governance dimensions

To better understand how ESG performance drives green innovation, ESG is disaggregated into its three core components

TABLE 15 Labor heterogeneity.

Variable	(1)	(2)
	GIN	GIN
ESG	0.0331*** (0.0069)	0.0635*** (0.0075)
GF	0.0432*** (0.0123)	0.0476** (0.0199)
ESG*GF	−0.0067 (0.0505)	0.0649 (0.0586)
Constant	−1.5785*** (0.5118)	−1.1816* (0.6123)
Control	YES	YES
FirmFE	YES	YES
YearFE	YES	YES
R ²	0.0368	0.0449

*, **, and *** denote statistical significance of parameters at the 10, 5, and 1%, respectively. The given values in parentheses present adjusted standard errors.

environmental (ESG_E), social (ESG_S), and governance (ESG_G) and their individual effects are assessed using separate two-way fixed effects models (Table 16).

The findings reveal a pronounced hierarchy of effects, with environmental performance exerting the strongest influence on green innovation, followed by governance and social performance, all significant at the 1% level. This pattern is consistent with (Liu and Jiang, 2025), who similarly found that the environmental sub-dimension dominates the ESG–green innovation relationship among Chinese A-share listed firms.

An important methodological clarification is warranted. Since ESG_E, ESG_S, and ESG_G are entered as independent sub-scores in separate models, the larger coefficient for ESG_E is not mechanically attributable to its weight in the Sino-Sec composite index but reflects a genuinely stronger marginal relationship with green innovation outcomes. Theoretically, the environmental dimension captures direct corporate actions environmental expenditures, green supply chain management, and pollution control that share an immediate causal link with green patent generation. By contrast, governance and social dimensions act as indirect facilitators: governance reduces agency costs and improves strategic oversight, while social performance builds stakeholder trust and enhances organizational learning, both supporting rather than directly driving innovation. Nonetheless, future research should validate these findings using alternative rating systems such as Bloomberg ESG or MSCI to confirm robustness across different weighting architectures.

7 Conclusion and policy implications

7.1 Conclusion

This study empirically examines the impact of corporate ESG performance on green innovation using panel data of Chinese A-share listed firms from 2010 to 2023. By applying two-way fixed effects, mediation, moderation, and instrumental variable approaches, the analysis provides robust evidence on the direct, indirect, and heterogeneous effects of ESG performance.

The results show that ESG performance significantly promotes green innovation, and this relationship remains stable across multiple robustness checks. Mechanism analysis indicates that ESG

TABLE 16 Influence of environmental, social, and governance factors on firms' green innovation.

Variable	(1)	(2)	(3)
	GIN	GIN	GIN
ESG_E	0.0533*** (0.00545)		
ESG_E		0.0034*** (0.0005)	
ESG_G			0.0053*** (0.0007)
Constant	−1.5109*** (0.4416)	−1.4814*** (0.4363)	−1.7345*** (0.4434)
Control	YES	YES	YES
FirmFE	YES	YES	YES
YearFE	YES	YES	YES
R ²	0.0358	0.0365	0.0369

*** denote statistical significance of parameters at the 1%. The given values in parentheses present adjusted standard errors.

enhances green innovation mainly by alleviating financing constraints and reducing information asymmetry, suggesting that ESG operates through substantive resource and information channels rather than symbolic signaling alone. In addition, green finance significantly strengthens this relationship, particularly in non-eastern regions where financial development is relatively weaker. Heterogeneity results further reveal that the positive impact of ESG is more pronounced in state-owned enterprises, eastern regions, and technology-intensive industries, reflecting stronger institutional support and resource endowments in these contexts.

7.2 Policy implications

Based on the empirical findings, several targeted policy implications are proposed.

First, policymakers should implement region-specific green finance policies. In central and western regions, governments should expand green credit supply by introducing interest subsidies for green projects, increasing policy-based financing, and encouraging financial institutions to develop ESG-linked loan products.

Second, regulatory authorities should strengthen ESG disclosure and enforcement mechanisms. For state-owned enterprises, ESG performance should be integrated into performance evaluation systems with clear and binding green R&D targets. For private firms, improving the standardization and credibility of third-party ESG ratings is essential to reduce information asymmetry and enhance investor confidence.

Third, in pollution-intensive and technology-intensive industries, governments should adopt targeted policy instruments, such as combining carbon trading mechanisms with innovation subsidies, tax incentives, and R&D support, to accelerate low-carbon technological transformation.

Fourth, financial institutions should embed ESG criteria into credit allocation decisions, offering preferential financing conditions to firms with strong ESG performance, thereby reinforcing the link between sustainability practices and innovation investment.

7.3 Limitations and future research

This study focuses on A-share listed firms, which provides a consistent and high-quality dataset within a well-defined

institutional context. Future research could build on this by incorporating non-listed firms and exploring broader firm populations to further enrich the understanding of ESG–innovation dynamics.

In addition, ESG performance in this study is measured using a widely recognized rating system. Subsequent studies may consider integrating multiple ESG rating sources or constructing composite indicators to capture ESG performance from different perspectives.

Future research could also explore cross-country comparisons and examine the dynamic evolution of ESG and green innovation over longer time horizons, thereby offering deeper insights into the role of institutional environments and temporal effects in shaping sustainable innovation outcomes.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

YZ: Writing – original draft, Data curation, Investigation, Conceptualization, Writing – review & editing, Software, Formal analysis, Methodology. SL: Writing – review & editing, Investigation, Software, Data curation, Methodology, Conceptualization, Visualization, Writing – original draft, Formal analysis. MT: Resources, Writing – review & editing, Funding acquisition, Conceptualization, Supervision, Methodology, Writing – original draft, Investigation.

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